# Classification Metrics

We will look at several metrics used to assess the quality of a given classifier.

1. Accuracy
2. Precision
3. Recall (also called *Sensitivity*)
4. F1 Score
5. Receiver Operating Characteristics

## REFERENCE

- Textbook Chapter 3, "Classification"
  - pp 107 - 119

## Why is Accuracy not enough?

- Recall that 'accuracy' is defined as

$$\frac{1}{\mathcal{N}_{test}} \sum_{i=1}^{\mathcal{N}_{test}} \mathbb{I}\{y_i = \hat{y}_i.\}$$

- Suppose my test set has 7,000 samples, of which
  - 50 have label '1'
  - 6,950 have label '0'
  - Such datasets are called *Unbalanced Datasets*
  - Sometimes, this situation is also referred to as *Data Imbalance*.
- Now, if I employ a naive classifier that outputs '0' for ALL inputs,
  - Accuracy = $\frac{6{,}950}{7000} \approx 99.29\%!!$

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# Classifier Outputs

- **True Positive (TP):** The model correctly predicted the positive class.
- **True Negative (TN):** The model correctly predicted the negative class.
- **False Positive (FP):** The model incorrectly predicted the positive class (Type I error).
- **False Negative (FN):** The model incorrectly predicted the negative class (Type II error).

In the context of the example discussed in Chapter 3 of the textbook (i.e., classifying a given MNIST image as **"5" or "not 5"**), the terms **True Positive (TP)**, **True Negative (TN)**, **False Positive (FP)**, and **False Negative (FN)** are defined as follows:

### True Positive (TP):

- **Definition**: The model correctly predicts an image as "5."
- **Example**: If the image is truly a "5" and the model predicts it as "5," it is a **true positive**.

### True Negative (TN):

- **Definition**: The model correctly predicts an image as "not 5."
- **Example**: If the image is truly **not a "5"** (e.g., a "3" or "7") and the model predicts it as "not 5," it is a **true negative**.

### False Positive (FP):

- **Definition**: The model incorrectly predicts an image as "5" when it is actually not "5."
- **Example**: If the image is truly a "7" (or any other digit except "5") but the model incorrectly predicts it as "5," it is a **false positive**. This is also known as a **Type I error**.

### False Negative (FN):

- **Definition**: The model incorrectly predicts an image as "not 5" when it is actually "5."
- **Example**: If the image is truly a "5" but the model incorrectly predicts it as "not 5," it is a **false negative**. This is also known as a **Type II error**.

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## Confusion Matrix

A **confusion matrix** is a table that summarizes the performance of a classification model by comparing predicted labels with true labels. For a binary classification problem (2-class), the confusion matrix is a 2x2 table with the following components:

| | **Predicted Positive** | **Predicted Negative** |
|---|---|---|
| **Actual Positive** | **True Positive (TP)** | **False Negative (FN)** |
| **Actual Negative** | **False Positive (FP)** | **True Negative (TN)** |

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## Metrics Derived from the Confusion Matrix

1. **Accuracy**: Measures the proportion of correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{TP + TN}{\text{Total Number of Test Samples}}$$

2. **Precision**: Proportion of true positive predictions among all positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. **Recall (Sensitivity)**: Proportion of actual positives identified correctly.

$$\text{Recall} = \frac{TP}{TP + FN}$$

4. **F1-Score**: Harmonic mean of precision and recall, balancing both.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

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- Going back to our earlier example, note that the confusion matrix is of the form | | **Predicted 1** | **Predicted 0** |

|---------------------|-----------------------|-----------------------| | **Actually 1** | TP=0| FN=50 | | **Actually 0** | FP=0| TN=6,950 |

- With this, we see that $\text{precision} = \text{recall} = 0$

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# Numerical Example

## Example Confusion Matrix

Suppose we have the following confusion matrix for a binary classification task:

| | **Predicted Positive** | **Predicted Negative** |
|---|---|---|
| **Actual Positive** | **50 (TP)** | **10 (FN)** |
| **Actual Negative** | **5 (FP)** | **100 (TN)** |

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1. **Accuracy**:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Substituting the values:

$$\text{Accuracy} = \frac{50 + 100}{50 + 100 + 5 + 10} = \frac{150}{165} \approx 0.909\,(90.9\%)$$

2. **Precision** (Positive Predictive Value):

$$\text{Precision} = \frac{TP}{TP + FP}$$

Substituting the values:

$$\text{Precision} = \frac{50}{50 + 5} = \frac{50}{55} \approx 0.909\,(90.9\%)$$

3. **Recall** (Sensitivity or True Positive Rate):

$$\text{Recall} = \frac{TP}{TP + FN}$$

Substituting the values:

$$\text{Recall} = \frac{50}{50 + 10} = \frac{50}{60} \approx 0.833\,(83.3\%)$$

4. **F1 Score** (Harmonic mean of Precision and Recall):

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Substituting the values:

$$\text{F1 Score} = 2 \times \frac{0.909 \times 0.833}{0.909 + 0.833} = 2 \times \frac{0.757}{1.742} \approx 0.869\,(86.9\%)$$

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## Summary: Why Accuracy Is Not Enough

While accuracy is an intuitive and widely used metric, it can be misleading in many situations, especially when dealing with **imbalanced datasets** (where one class significantly outnumbers the other). Here are some key reasons:

1. **Imbalanced Classes**:
   - If the dataset contains 95% negatives and 5% positives, a model predicting all instances as negative will achieve 95% accuracy but will completely fail to identify the positive class.
2. **False Positives and False Negatives**:
   - High accuracy does not distinguish between the costs of false positives (e.g., predicting a healthy patient as sick) and false negatives (e.g., missing a cancer diagnosis).

- In applications like medical diagnosis, fraud detection, or spam filtering, false negatives or positives may have significantly different impacts.

3. **Misleading in Multi-Class Problems**:

   - In multi-class scenarios, high accuracy might be due to the model performing well on the majority class while failing on minority classes.

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